

An Open-source Toolkit for Seismic Vulnerability Assessment

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ABSTRACT

This study introduces an open-source numerical modeling toolkit and library developed to assess the seismic vulnerability of building structures. Using an object-oriented approach in OpenSeesPy, the toolkit seamlessly creates an idealized model of building classes as multi-degree-of-freedom oscillators, accurately capturing the complex dynamic behavior of various structural types. The library supports linear (i.e., gravity and modal analyses), nonlinear static analyses (including static and cyclic pushover analyses) and dynamic analyses (i.e., cloud analyses), facilitating a thorough evaluation of seismic performance. The toolkit generates probabilistic demand-intensity models through robust regression of cloud analysis results, linking seismic demand to building response, and derives fragility functions that estimate the probability of exceeding various damage states under increasing levels of ground shaking. Furthermore, it produces vulnerability functions to quantify the overall risk and expected losses for building classes. By offering an accessible and flexible library for seismic vulnerability assessment, this toolkit improves the ability to evaluate and reduce earthquake-related damage across a wide array of structural designs. Its open-source nature fosters transparency, reproducibility, and ongoing development within the global earthquake engineering community. The toolkit is maintained by the Global Earthquake Model (GEM) Foundation on GitHub, and benefits from continuous updates and enhancements contributed by experts and researchers worldwide.